

Machine Learning Based Breast Cancer Visualization and Classification

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Abstract— In contemporary years, the categorization of breast cancer has become an engrossing subject in the department of healthcare informatics due to prodigious deaths of the women across the world caused by this cancer. With the upcoming heed and variety of approaches in image processing and machine learning (ML), there has been an endeavor to erect a pattern recognition model that is well-grounded to boost the diagnosis standard. Diverse research has been attempted on mastering the prediction of the possibility of breast cancer using predefined data mining algorithms. In this paper, a model is presented using the support vector machine (SVM) algorithm for the manual categorizing of the histology images of breast cancer samples into benign and malignant subclasses to anticipate the interpretation. Primarily all the data incorporating a set of 30 features relating to the cell nuclei shown in the digitalized images of fine needle aspirate (FNA) of a breast mass are considered. Ten existing values of features are added up for every nuclei sample then the mean, the standard deviation, the worst and largest of the mentioned attributes are measured proceeding to 30 features. The total features obtained are visualized and apprehended to gain insight for future diagnosis. The principal component analysis (PCA) dimensionality reduction strategy is implemented to successfully augment the valiance of the attributes resolving eigenvector problem. The ultimate outcome is conceptualized using the confusion matrix and the receiver operating characteristic curve (ROC). This SVM forged model proves to show 97% accuracy with the recommended dataset.

Keywords—breast cancer, classing, confusion matrix, principal component analysis, receiver operating characteristic curve, support vector machine, visualization.

I. INTRODUCTION

As started by the American Cancer Society, there are 276,480 estimated instances of breast cancer in women. Out of which 42,170 resulted in death in the USA for 2020 [10]. In Arab countries, 30 years age women are spotted more vulnerable to this type of cancer. Many supervised machine learning algorithms handled this prediction. The NB6, C4.57, KNN8 are contrasted by various researches previously. Breast cancer transpires as the thickening of the cells in breast tissue. magnetic resonance imaging (MRI), mammogram, ultrasound perceives the existence of breast cancer. Fine needle aspirate (FNA) predicts breast cancer results. Early detection of cancer reduces cost and the success rate of treatment doubles.

The principal component analysis is the linear procedure that affiliates feature extraction and feature selection, in which, high dimensional data is transformed into low forming smaller subsets and tend to find interconnection among the variables. The SVM is a superintended classing technique to perceive optimal hyper-plane and builds into two sets, then elects the category in which the newly arrived value should be placed. SVM facilitates complex decision partitions. This proposed model is built using SVM as the data in the dataset is non-linear data with different dimensions. So, this algorithm has a sophisticated way of transforming the non-

linear data to linear data to suitably fit the pertinent model. SVM can be applied in various sectors i.e, health, satellite data, cancer data, etc [1]. Random split algorithm can also serve the purpose but it creates individual decision trees for each attribute and then the combined outputs from these trees is taken and a final prediction is made. This job is tedious compared to SVM based modelling and also SVM popularity is high classification accuracy and robustness compared to other classifiers [4].

Many researchers have published works based on the ground study in this area of interest, taking into consideration various parameters and metrics in their projects. Shang Gao et al [2] work overcomes the complications of the long-established ways of breast cancer prognosis using credit gauging system by inspecting various kernel calculations in SVM. Ilias Maglogiannis [3] used a relative swotting, comparing the three techniques, Bayesian Classifiers, Neural Networks, SVM. From which, the SVM showed superiority among them by metrics such as sensitivity, accuracy, specificity. Vikas Chaurasia [5] objective was to get the variations among the two most favoured algorithms. Naive Bayes algorithm and J48 algorithm is featured in the diagnosis of cancer within the large dataset comprising of 683 cases. The final results manifest Naive Bayes algorithm to be the perfect predictor with 97% accuracy and J48 algorithm with 93% precision. Naresh Khuriwal [6] preferred the Wisconsin dataset curbing the multiple samples of breast cancer patients. The Logistic, artificial neural network algorithms give the best option with 97% correctness. Labeed K Abdulgafoor [7] used wavelet change and K means clustering form for prediction. S. Padma Priya [8] have done survey work that intensifies the breast cancer discovery with ID3 and C4.5 splitting algorithms.

This paper projects the usage of the support vector machine algorithm as the classification algorithm to cleave the samples of cell nuclei into benign (B) and malignant (M) depending on the feature scrutiny. The datasets contain 569 samples of malignant and benign tumor cells. The values are designated as 1 = Malignant (Cancerous) – Present, 0 = Benign (Not Cancerous) – Absent. Descriptive statistics recapitulate the characteristics of the entire data. Count, mean, standard deviation are few metrics. Histograms and density plots are utilized for multimodal visualization and as a mode of picturing all the features.

II. METHODOLOGY

A. Model Data Flow

The flowchart represents the approaches at every step of model building. The initial step includes data gathering and observing the type of attributes listed in dataset. The next step constitutes statistical as well as the diagrammatic depiction of the values in the shape of histograms and density plots for mean, standard deviation and worst, to understand the relation between them. The feature standardization and feature

decomposition play a crucial role in the correctness of the suggested method. Then the SVM algorithm is garrisoned for stratifying training and test samples into benign and malignant cases. Fig. 1 gives a comprehensive view of the theory initiated.

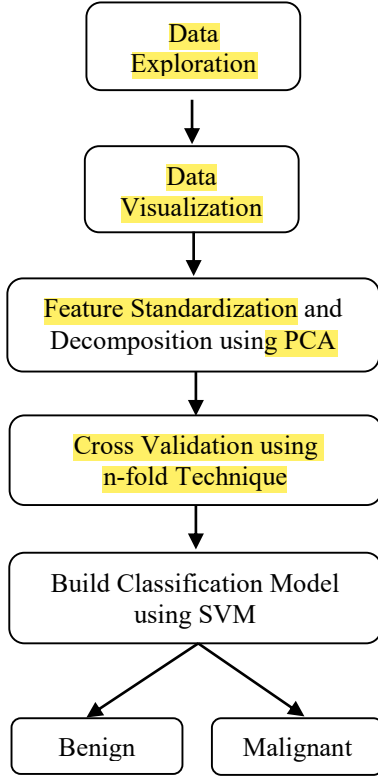


Fig. 1. Flowchart depicting diagnosis

In this section, the in-depth knowledge for grooming the model with the suitable steps is demonstrated. Machine learning algorithms are pre-written codes that are easily lodged to memorize the patterns from the training set given before and seek results from the testing dataset. The training procedure continues until the preferred output is acquired. Classification is carried out as a data mining function that sorts all the sets of data appropriately among the two categories i.e., benign and malignant using 70% data as a trainset and 30% as a testset. The steps for constructing the predictive model are discussed in sequence.

B. Identification of the dataset

The Wisconsin breast cancer dataset projects the features severely based on fine needle aspirate (FNA) test. The breast cancer Wisconsin (diagnostic) data (WDBC) set is made up of 569 samples of nuclei of the cell. It constitutes 32 features. Two attributes include the id of the samples and the other is types of cancer (M and B). The leftover 30 features are enumerated by calculated mean, standard deviation, worst of the predefined ten attributes given in Table 1 for each cell nuclei [9].

After finding all the attributes, inspect the data to get the type of data. All the variables are of float type excluding id which is of int type and diagnosis which is of the object type. Fig. 2 shows all the features utilized in this study.

C. Exploratory Data Analysis- Visualization

The exploratory data analysis (EDA) proceeds with taking a closer look at data by comprehending the structure of data, apportioning, and finding the interrelationship among the values to gain insight. Before modelling any machine learning model, this step of feature engineering is implemented to highlight salient information. Two approaches are applied in this project: descriptive statistics, unimodal, and multimodal visualization.

1) *Descriptive Statistics*: It embraces the statistics that condenses the entire characteristics of data into a few numeric metrics such as mean, count, standard deviation, etc [11]. Fig. 3. gives an account of statistical knowledge about the key features.

TABLE I. STANDARD CHARACTERISTICS OF BREAST CELL NUCLEI

S. No.	Standard Characteristics of breast cell nuclei
1.	radius
2.	texture – standard deviation of greyscale values
3.	perimeter
4.	area
5.	smoothness- variation in radius length
6.	compactness- ((perimeter) ² /area)-1
7.	concavity – the severity of concave portions on contours
8.	concave points- no of concave portions on contour
9.	Symmetry - fractal dimension- (costal approx.-1)

Data columns (total 31 columns):

#	Column	Non-Null	Count	Dtype
0	diagnosis	569	non-null	object
1	radius_mean	569	non-null	float64
2	texture_mean	569	non-null	float64
3	perimeter_mean	569	non-null	float64
4	area_mean	569	non-null	float64
5	smoothness_mean	569	non-null	float64
6	compactness_mean	569	non-null	float64
7	concavity_mean	569	non-null	float64
8	concave points_mean	569	non-null	float64
9	symmetry_mean	569	non-null	float64
10	fractal_dimension_mean	569	non-null	float64
11	radius_se	569	non-null	float64
12	texture_se	569	non-null	float64
13	perimeter_se	569	non-null	float64
14	area_se	569	non-null	float64
15	smoothness_se	569	non-null	float64
16	compactness_se	569	non-null	float64
17	concavity_se	569	non-null	float64
18	concave points_se	569	non-null	float64
19	symmetry_se	569	non-null	float64
20	fractal_dimension_se	569	non-null	float64
21	radius_worst	569	non-null	float64
22	texture_worst	569	non-null	float64
23	perimeter_worst	569	non-null	float64
24	area_worst	569	non-null	float64
25	smoothness_worst	569	non-null	float64
26	compactness_worst	569	non-null	float64
27	concavity_worst	569	non-null	float64
28	concave points_worst	569	non-null	float64
29	symmetry_worst	569	non-null	float64
30	fractal_dimension_worst	569	non-null	float64

dtypes: float64(30), object(1)

Fig. 2. Attributes of WDBC

2) *Unimodal and Multimodal Visualization*: Visualization forecasts cartesian space into abstract images. Mean, standard error (SE), and worst for the primary 10 features of the WDBC are projected through unimodal representations like histograms and density plots. Fig. 4. describes the histogram mean designation of area, compactness, concave points

features. Fig. 5. describes the histogram mean designation of remaining. Fig. 6. and 7. spots density plots for all the means calculated. Furthermore, histogram and density plot SE, worst suffixes can also be gauged. Whereas, multimodal is done through a correlation matrix that gives the interaction of the strength offsetting the weaker ones. This matrix organization formats two-dimensional space for the mean values of

parameters along both axes. The range for strong positive tie-up is 1 to 0.75 and moderate correlation ranges between 0.5 to 0.75. There is a possibility of negative ranges that manifests the least relationship. Fig. 8 provides the breast cancer feature matrix.

```
In [13]: #basic descriptive statistics
data.describe()
```

Out[13]:

	radius_mean	texture_mean	perimeter_mean	area_mean	smoothness_mean	compactness_mean	concavity_mean	concave points_mean	symmetry_mean	fract
count	569.000000	569.000000	569.000000	569.000000	569.000000	569.000000	569.000000	569.000000	569.000000	
mean	14.127292	19.289649	91.969033	654.889104	0.096360	0.104341	0.088799	0.048919	0.181162	
std	3.524049	4.301036	24.298981	351.914129	0.014064	0.052813	0.079720	0.038803	0.027414	
min	6.981000	9.710000	43.790000	143.500000	0.052630	0.019380	0.000000	0.000000	0.106000	
25%	11.700000	16.170000	75.170000	420.300000	0.086370	0.064920	0.029560	0.020310	0.161900	
50%	13.370000	18.840000	86.240000	551.100000	0.095870	0.092630	0.061540	0.033500	0.179200	
75%	15.780000	21.800000	104.100000	782.700000	0.105300	0.130400	0.130700	0.074000	0.195700	
max	28.110000	39.280000	188.500000	2501.000000	0.163400	0.345400	0.426800	0.201200	0.304000	

8 rows × 30 columns

Fig. 3. Sample Screenshot of Descriptive statistics

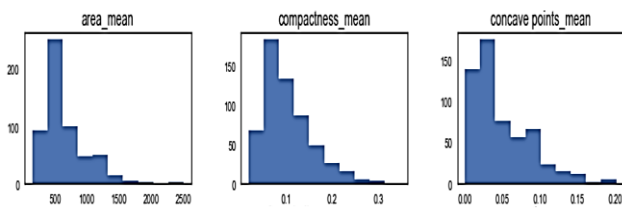


Fig. 4. Histogram mean suffix for area, compactness, concave points

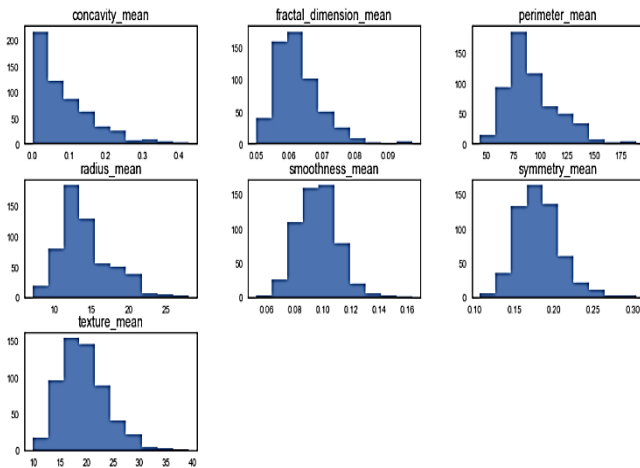


Fig. 5. Histogram mean suffix for are, compactness, concave points

From Fig. 4. and Fig. 5. we can expect that concavity, the concave points attributes have an exponential distribution of input variables, and texture, smoothness, symmetry attributes have a Gaussian distribution which is mostly assumed by the machine learning trainers. As seen from Fig. 6. And Fig. 7. perimeter, radius, area, concavity, compactness aspects show an exponential distribution, while the texture, smooth, symmetry attributes predicts virtually as Gaussian distribution.

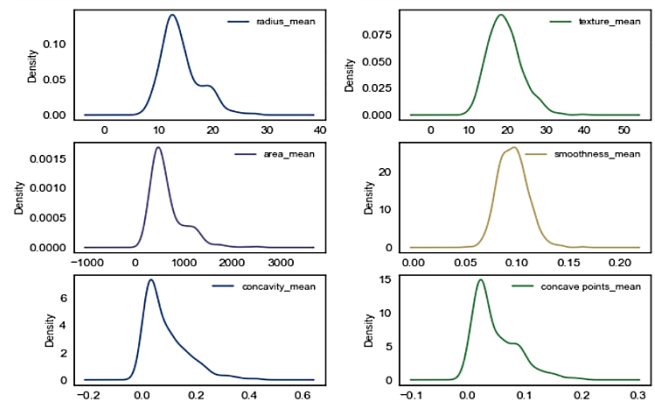


Fig. 6. Density plot mean suffix for radius, texture, area, smoothness, concavity, concave points attributes.

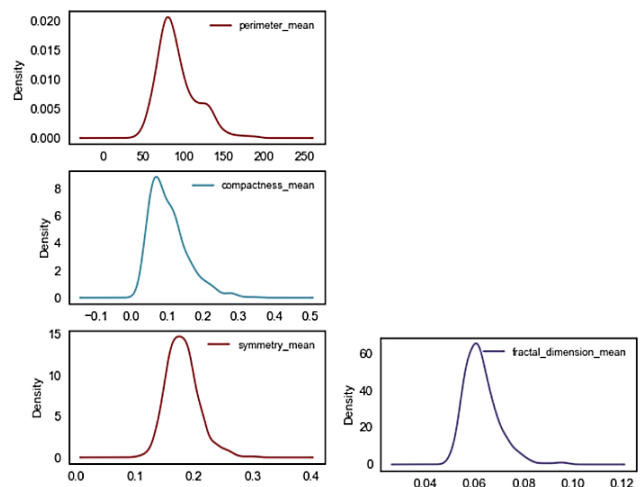


Fig. 7. Density plot mean suffix for perimeter, compactness, symmetry, fractal_dimension attributes

The mean area has a strong positive interdependency with mean values of radius and perimeter. Additionally, some parameters that are moderately positive correlated are concavity and area, concavity and perimeter, etc. Besides, we come across a strong negative correlation between fractal dimension with radius, texture, perimeter mean values. To sum up, we culminate that the mean values of cell radius, perimeter, area, compactness, concavity, and concave points are vastly preferred in the classification of breast cancer. Larger values of these parameters tend to show a correlation with malignant type of tumours. The rest of the features do not preferably brace to prognosis.

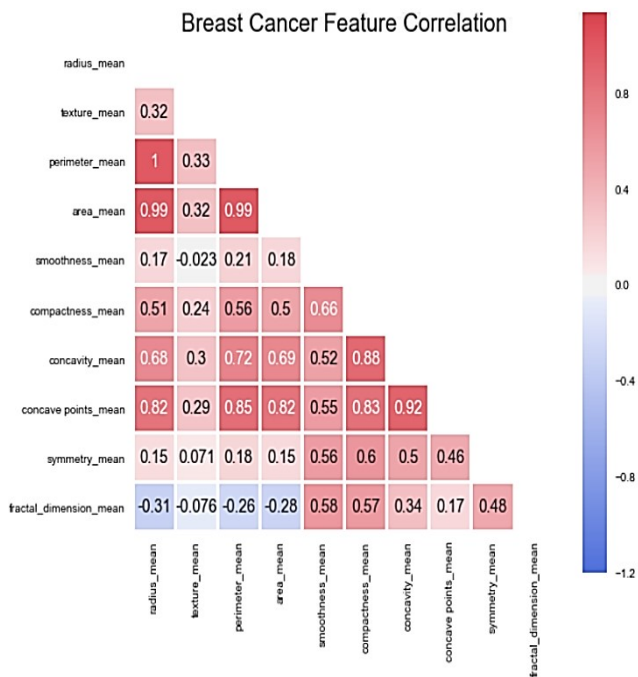


Fig. 8. Feature correlation matrix for the mean of features

D. Processing the data

It is a vital step in building the machine learning model to fit the model to its requirements. The series of activities occurring in this step encompasses encoding, feature standardization, and feature decomposition. The label encoder transforms all the categorical type variables into numerical ones. Diagnosis features with class label M and B are encoded into integers 1 and 0. The presence of benign tumor cells designates 0 and malignant as 1.

1) Feature Standardization: Here, the attributes are with Gaussian distribution of differing means and variance i.e features are not on matching scale, so in this step they are remodeled to standard Gaussian distribution of mean-0 and standard deviation-1. The word "data" is plural, not singular.

2) Feature decomposition: Principal Component Analysis (PCA) is a widely favored dimensionality reduction method to work with only X and Y dimensions covering all essential features without any loss of valuable data. In Fig. 9.

we chose to apply converting the 3d to 2d space classifying all the samples according to their class type.

By plotting the screen plot for eigenvalues from PCA we get to know how many principal components to preserve. Fig. 10. shows the plotting. At two components the slope of the curve deteriorates assuming to take three components into deliberation.

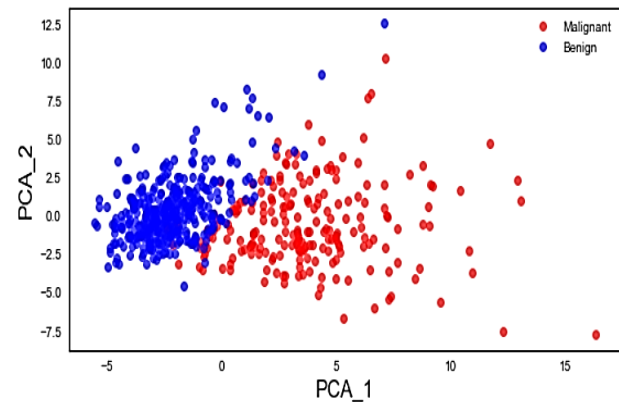


Fig. 9. Feature decomposition by PCA

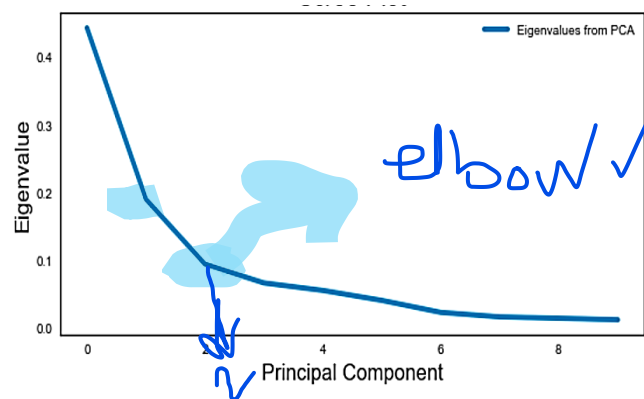


Fig. 10. Screen plot to gain principal components

E. Predictive Modelling using SVM

Initially, in this phase, we split the breast cancer dataset into test and training tuples. 70% of data to inculcate the model to get familiar with the structure. The rest of the 30% values are assumed as test data to model the predictor. Fig. 11. tells that from 569 samples, 423 cases are regarded as a train set and 146 as a test set.

The SVM's are popular and have a distinguished way of transforming and executing ably on a large number of databases effectively.

Fig 12. gives the classifier score of prediction as 0.953 which is approximately 95% accuracy when the whole set of features are taken into account and in Fig. 13 the best three components after the feature selection strategy, are taken into consideration thereby advancing the score to 0.97 which is 97% accuracy. We can finally deduce that feature selection is a crucial step in elevating the model performance.


```
In [4]: from sklearn.model_selection import train_test_split

        ##Split data set in train 70% and test 30%
        X_train, X_test, y_train, y_test = train_test_split( X, y, test_size=0.25, random_state=7)
        X_train.shape, y_train.shape, X_test.shape, y_test.shape

Out[4]: ((426, 30), (426,), (143, 30), (143,))
```

Fig. 11. Screen plot to gain principal components

```
In [6]: # 5. Divide records in training and testing sets.
        X_train, X_test, y_train, y_test = train_test_split(Xs, y, test_size=0.3, random_state=2, stratify=y)

        # 6. Create an SVM classifier and train it on 70% of the data set.
        clf = SVC(probability=True)
        clf.fit(X_train, y_train)

        #7. Analyze accuracy of predictions on 30% of the holdout test sample.
        classifier_score = clf.score(X_test, y_test)
        print (classifier_score)

0.9532163742690059
```

Fig. 12. Accuracy before feature selection

```
In [11]: from sklearn.feature_selection import SelectKBest, f_regression
        clf2 = make_pipeline(SelectKBest(f_regression, k=3),SVC(probability=True))

        scores = cross_val_score(clf2, Xs, y, cv=3)

        # Get average of 3-fold cross-validation score using an SVC estimator.
        n_folds = 3
        cv_error = np.average(cross_val_score(SVC(), Xs, y, cv=n_folds))
        print '\nThe {}-fold cross-validation accuracy score for this classifier is {:.2f}\n'.format(n_folds, c
```

The 3-fold cross-validation accuracy score for this classifier is 0.97

Fig. 13. Accuracy after feature selection

III. RESULTS

The confusion matrix and receiver operating characteristic curve (ROC) is drawn by including all the features of the set, to cumulate the final judgment. Fig. 14. outlines a confusion matrix and Fig. 15. shows a table with precision, support, fl-score metrics. From this matrix, we can retrieve the values: true positive (TP) = 106, true negative (TN)= 57, false positive (FP) = 1, false negative (FN) = 7.

Similarly, the following presumptions can also made:

- A total of 174 cases of the presence of cancer are detected.
- Out of all the predicted ones, 58 are observed to be malignant and 113 are benign by the classifier model.
- But factually 107 members are suffering from malignant and 64 with benign type of tumours.

	0	1
Actual Values		
0	106	1
1	7	57
	Predicted Values	

Fig. 14. Resulting Confusion matrix

	precision	recall	f1-score	support
0	0.94	0.99	0.96	107
1	0.98	0.89	0.93	64
avg / total	0.95	0.95	0.95	171

Fig. 15. Summary of precision recall, f1-score for each class

The above metrics are computed by the equations below:

$$\text{Precision} = \text{TP}/(\text{TP}+\text{FP}) = 57/58 = 0.98 \quad (1)$$

$$\text{Recall} = \text{TP}/(\text{TP}+\text{FN}) = 57/64 = 0.89 \quad (2)$$

$$\begin{aligned} \text{F1-score} &= 2 * (\text{Precision} * \text{Recall}) / (\text{Precision} + \text{Recall}) \quad (3) \\ &= 2 * (0.8722) / 1.87 = 0.93 \end{aligned}$$

In ML, another electable performance metric is the receiver operating characteristic (ROC) curve. On X-axes of the curve is the false positive rate (FPR) and Y-axes is the true positive rate (TPR) estimated using the equations (1) and (2):

True positive rate (or Recall):

$$\text{TPR} = \text{TP} / (\text{TP} + \text{FN}) = 57/64 = 0.89 \quad (4)$$

False positive rate:

$$\text{FPR} = \text{FP} / (\text{FP} + \text{TN}) = 1/107 = 0.01 \quad (5)$$

Every prediction part of the confusion matrix is plotted as a single point in ROC dimensional scope. The diagonal line is the chance of random points and plots above the line are perfect results of splitting and below are the worst ones as in Fig. 16.

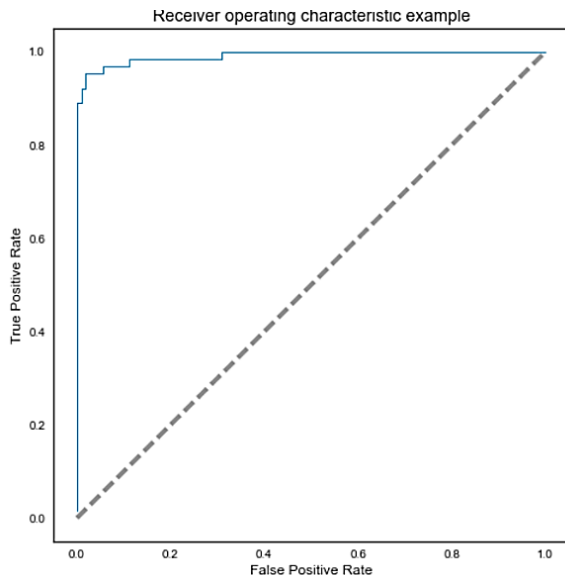


Fig. 16. ROC curve for SVM

IV. CONCLUSION

This paper presented strives to primarily detect the tumour cells from the breast mass giving precise and speedy confirmation. This reduces the chance of severity and enhances the survival rate with a wide range of treatment

alternatives. The main goal of this model is to classify the type of cancer specifically, so the SVM classifier is used as a modeling algorithm. Others such as random forest, decision trees can also be opted. Previously innumerable number of authors have done over a lot of research with ID3, C4.5 but using the kernel SVM intensifies the model precision and correctness.

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